

22MAT230 - MATHEMATICS FOR COMPUTING 4

ADAPTIVE TOEPLITZ CONVOLUTION FOR ECG SIGNAL ANALYSIS

GROUP - C17:

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TITLE :

Adaptive Toeplitz Convolution for ECG Signal Analysis

Electrocardiogram (ECG) signals are widely used to monitor the electrical activity of the heart and to detect cardiac abnormalities. Analyzing ECG signals is important for identifying irregular heart conditions such as arrhythmia and other cardiovascular disorders. However, ECG signals often contain complex patterns and variations that make automated analysis challenging. Signal processing techniques can help extract meaningful information from ECG data and improve the detection of abnormal patterns.

In this project, an adaptive Toeplitz convolution approach is applied to ECG signals to enhance important variations in the signal. The convolution is performed using a Toeplitz matrix representation with a kernel that adapts based on the statistical properties of the signal. After processing the signals, several statistical features are extracted and used to train a machine learning classifier. The classifier then categorizes ECG signals into normal and abnormal classes, allowing effective automated analysis of heart signal patterns.

INTRODUCTION :

- Electrocardiogram (ECG) signals are widely used in medical diagnostics to monitor the electrical activity of the heart. Detecting abnormalities in ECG signals is important for diagnosing heart diseases such as arrhythmia and cardiac irregularities.
- Traditional ECG analysis requires manual inspection by medical experts, which can be time consuming and subjective. Signal processing and machine learning techniques provide automated solutions for ECG classification.
- In this project, an adaptive Toeplitz convolution approach is used to enhance ECG signal patterns and extract meaningful features. These features are used to train a classifier that can distinguish between normal and abnormal ECG signals.

DATASET DESCRIPTION :

The dataset used in this project is the **ECG5000 dataset**, which contains ECG signals collected from medical recordings.

Dataset characteristics:

- Total signals: **5000**
- Samples per signal: **140**
- Each signal represents one heartbeat cycle
- Original dataset contains **5 classes**
- For simplicity, classes are converted to **binary classification**

Class labels used in this project:

- **0** → **Normal ECG**
- **1** → **Abnormal ECG**

This dataset provides sufficient variation in heart signal patterns to evaluate signal processing and classification techniques.

PROBLEM STATEMENT :

ECG signals contain complex patterns that can vary significantly between healthy and abnormal heart conditions.

The main challenges include:

- Identifying abnormal patterns in ECG signals
- Extracting meaningful signal features
- Developing an efficient method for classification

The goal of this project is to detect abnormalities in ECG signals using signal processing techniques combined with machine learning.

OBJECTIVES :

The objectives of this project are:

- To analyze ECG signals using signal processing techniques
- To implement adaptive Toeplitz convolution for signal enhancement
- To extract statistical features from processed ECG signals
- To classify ECG signals into normal and abnormal categories
- To evaluate the classification performance using accuracy and ROC analysis

METHODOLOGY :

The overall workflow of the project is as follows:

- Load ECG dataset
- Preprocess the ECG signals
- Apply adaptive Toeplitz convolution
- Extract statistical features
- Split dataset into training and testing sets

- Train KNN classifier
- Evaluate model performance
- Visualize ECG signals and classification results

ADAPTIVE TOEPLITZ CONVOLUTION :

Toeplitz matrices are special matrices where each diagonal has constant values. They are widely used in signal processing for implementing convolution operations efficiently.

In this project, Toeplitz matrices are used to perform convolution on ECG signals. The convolution kernel is designed adaptively based on signal statistics.

Adaptive kernel size: $k = 2 \lfloor \alpha \sigma + \beta \log(n) \rfloor + 1$

where

- σ = standard deviation of signal
- n = signal length
- α and β = tuning parameters

Kernel values are computed as: $K(i) = \mu + (i - c)\sigma$

where

- μ = mean of signal
- σ = standard deviation
- c = center index of kernel

This adaptive kernel helps highlight important signal variations and abnormalities.

FEATURE EXTRACTION :

After applying Toeplitz convolution, statistical features are extracted from the processed signal.

The extracted features include:

1. **Mean**
2. **Standard deviation**
3. **Signal range (max – min)**
4. **Total variation**
5. **Skewness**
6. **Kurtosis**
7. **Root Mean Square (RMS)**

These features summarize the signal characteristics and are used as inputs for the machine learning classifier.

CLASSIFICATION :

A **K-Nearest Neighbors (KNN)** classifier is used for ECG classification.

KNN works by comparing feature vectors of signals and assigning a class based on the majority class among the nearest neighbors.

Steps involved:

- Extract feature vectors from ECG signals
- Split dataset into training and testing sets
- Train the KNN classifier using training data
- Predict classes for testing data
- Evaluate classification accuracy

PERFORMANCE EVALUATION :

The performance of the classification model is evaluated using:

Accuracy

Measures the percentage of correctly classified signals.

Confusion Matrix

Shows classification results for:

- True Positive
- True Negative
- False Positive
- False Negative

ROC Curve

The ROC curve represents the relationship between:

- True Positive Rate
- False Positive Rate

The Area Under the Curve (AUC) indicates the overall performance of the classifier.

Higher AUC values indicate better classification performance.

VISUALIZATION :

Several visualizations are used to understand the signal behavior and classification results.

These include:

- Normal ECG signal
- Abnormal ECG signal
- Adaptive Toeplitz convolution output
- ROC curve
- Heatmap representation of ECG signals

These visualizations help analyze differences between normal and abnormal ECG patterns.

MATLAB CODE :

```
tic

clc; clear; close all;

%% -----
% 1. Load ECG5000 Dataset
%% -----

trainData = load('ECG5000_TRAIN.txt');
testData  = load('ECG5000_TEST.txt');

data = [trainData; testData];

labels = data(:,1);
signals = data(:,2:end);

[numSamples, n] = size(signals);

%% -----
% 2. Convert to Binary (Normal = 0, Abnormal = 1)
%% -----

labels_binary = zeros(size(labels));
labels_binary(labels ~= 1) = 1;

%% -----
% 3. Normalize Signals
%% -----

signals = normalize(signals,2);

%% -----
% 4. Adaptive Toeplitz Convolution Function
%% -----

function y = adaptive_toeplitz(x)

    n = length(x);

    mu = mean(x);
    sigma = std(x);

    alpha = 2;
    beta = 0.8;
```

```

k = 2 * floor(alpha*sigma + beta*log(n)) + 1;
k = max(k,5);
k = min(k,15);

center = (k+1)/2;
K = zeros(1,k);

for i = 1:k
    K(i) = mu + (i-center)*sigma;
end

first_col = [x(1), zeros(1,k-1)]';
first_row = x;

T = toeplitz(first_col, first_row);

y = K * T;
y = y(1:n);

end

%% -----
% 5. Feature Extraction
%% -----

features_toeplitz = zeros(numSamples,7);
features_standard = zeros(numSamples,7);

for i = 1:numSamples

    x = signals(i,:);

    % Toeplitz convolution
    y1 = adaptive_toeplitz(x);

    % Standard convolution
    y2 = conv(x, ones(1,5)/5, 'same');

    % Toeplitz features
    features_toeplitz(i,1) = mean(y1);
    features_toeplitz(i,2) = std(y1);
    features_toeplitz(i,3) = max(y1)-min(y1);
    features_toeplitz(i,4) = sum(abs(diff(y1)));
    features_toeplitz(i,5) = skewness(y1);
    features_toeplitz(i,6) = kurtosis(y1);
    features_toeplitz(i,7) = rms(y1);

    % Standard convolution features
    features_standard(i,1) = mean(y2);

```



```

features_standard(i,2) = std(y2);
features_standard(i,3) = max(y2)-min(y2);
features_standard(i,4) = sum(abs(diff(y2)));
features_standard(i,5) = skewness(y2);
features_standard(i,6) = kurtosis(y2);
features_standard(i,7) = rms(y2);

end

%% -----
% 6. Train-Test Split
%% -----

cv = cvpartition(labels_binary,'HoldOut',0.3);

X_train_T = features_toeplitz(training(cv),:);
Y_train    = labels_binary(training(cv));

X_test_T   = features_toeplitz(test(cv),:);
Y_test     = labels_binary(test(cv));

X_train_S = features_standard(training(cv),:);
X_test_S  = features_standard(test(cv),:);

%% -----
% 7. Train KNN (Toeplitz)
%% -----

Mdl_T = fitcknn(X_train_T,Y_train,'NumNeighbors',5);
Y_pred_T = predict(Mdl_T,X_test_T);

acc_T = sum(Y_pred_T == Y_test)/length(Y_test);
conf_T = confusionmat(Y_test,Y_pred_T);

%% -----
% 8. Train KNN (Standard Convolution)
%% -----

Mdl_S = fitcknn(X_train_S,Y_train,'NumNeighbors',5);
Y_pred_S = predict(Mdl_S,X_test_S);

acc_S = sum(Y_pred_S == Y_test)/length(Y_test);
conf_S = confusionmat(Y_test,Y_pred_S);

%% -----
% 9. Display Results
%% -----

disp('===== TOEPLITZ RESULTS =====')
disp(['Accuracy: ',num2str(acc_T)])

```

```

disp('Confusion Matrix:')
disp(conf_T)

disp('===== STANDARD RESULTS =====')
disp(['Accuracy: ',num2str(acc_S)])
disp('Confusion Matrix:')
disp(conf_S)

%% -----
% 10. ROC Curve
%% -----

[~,score_T] = predict(Mdl_T,X_test_T);

[Xroc,Yroc,~,AUC] = perfcurve(Y_test,score_T(:,2),1);

figure
plot(Xroc,Yroc,'LineWidth',2)
xlabel('False Positive Rate')
ylabel('True Positive Rate')
title(['ROC Curve (AUC = ',num2str(AUC),')'])
grid on

%% -----
% 11. Example Signals
%% -----

figure('Color','w')

normal_idx = find(labels_binary==0,1);
abnormal_idx = find(labels_binary==1,1);

subplot(3,1,1)
plot(signals(normal_idx,:), 'b')
title('Normal ECG (Normalized)')
grid on

subplot(3,1,2)
plot(signals(abnormal_idx,:), 'r')
title('Abnormal ECG (Normalized)')
grid on

subplot(3,1,3)
plot(adaptive_toeplitz(signals(abnormal_idx,:)), 'k')
title('Adaptive Toeplitz Output')
grid on

%% -----
% 12. Heatmap of All 5000 ECG Signals
%% -----

```

```

figure
imagesc(signals)

colorbar
xlabel('Sample Index')
ylabel('Signal Number')
title('Heatmap of All ECG Signals (5000)')
colormap(jet)

%% -----
% 13. Heatmap of Normal ECG Signals
%% -----

figure
imagesc(signals(labels_binary==0,:))

colorbar
xlabel('Sample Index')
ylabel('Normal Signal Number')
title('Heatmap of Normal ECG Signals')
colormap(jet)

%% -----
% 14. Heatmap of Abnormal ECG Signals
%% -----

figure
imagesc(signals(labels_binary==1,:))

colorbar
xlabel('Sample Index')
ylabel('Abnormal Signal Number')
title('Heatmap of Abnormal ECG Signals')
colormap(jet)

toc

```

RUN TIME : 3.998049 seconds

RESULTS :

===== TOEPLITZ RESULTS =====

Accuracy: 0.958

Confusion Matrix:

851	25
-----	----

38	586
----	-----

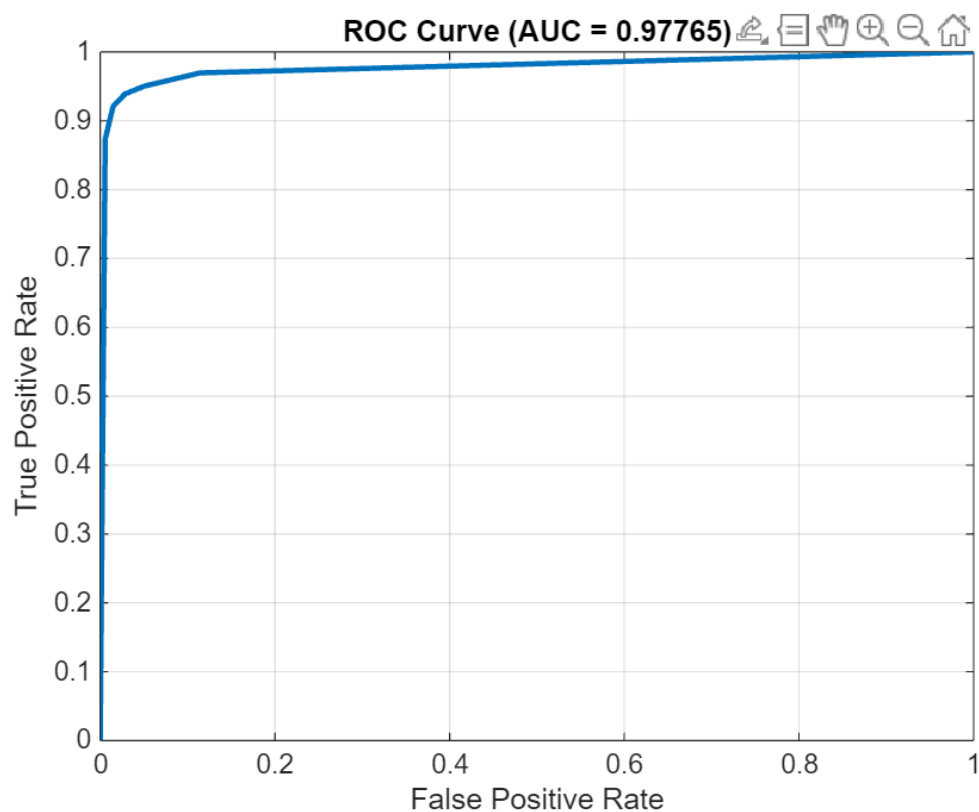
===== STANDARD RESULTS =====

Accuracy: 0.964

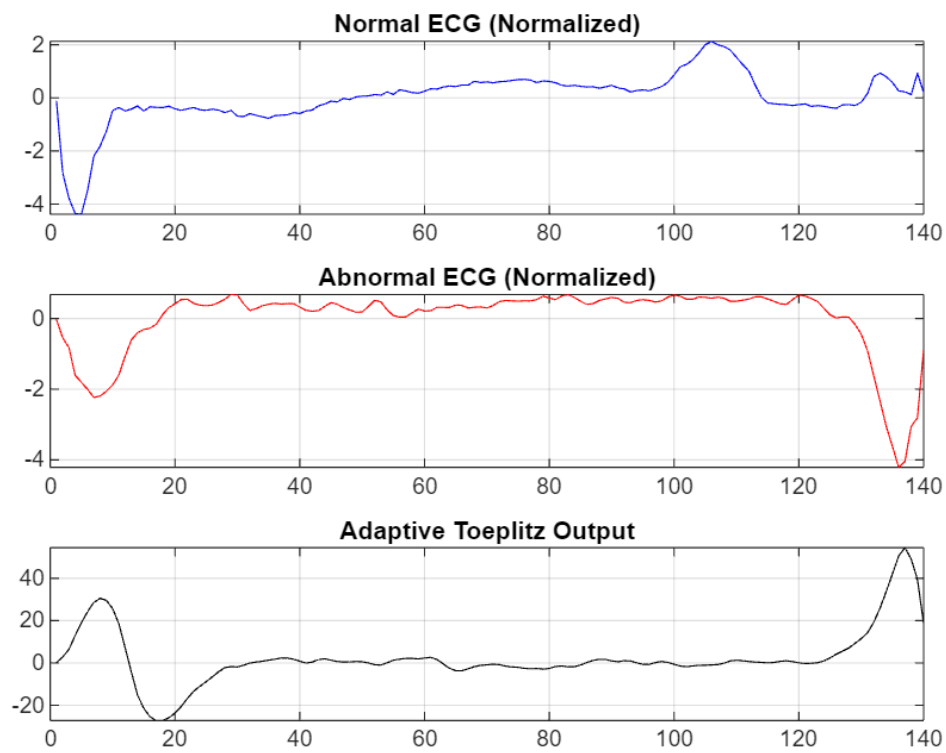
Confusion Matrix:

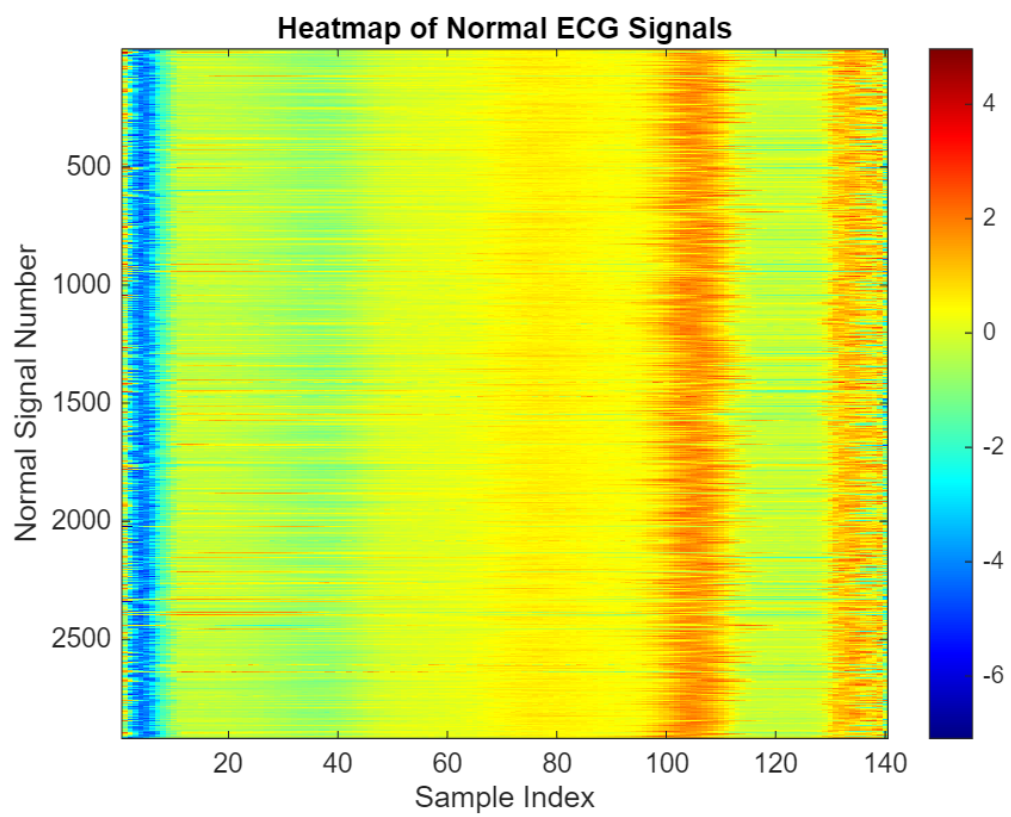
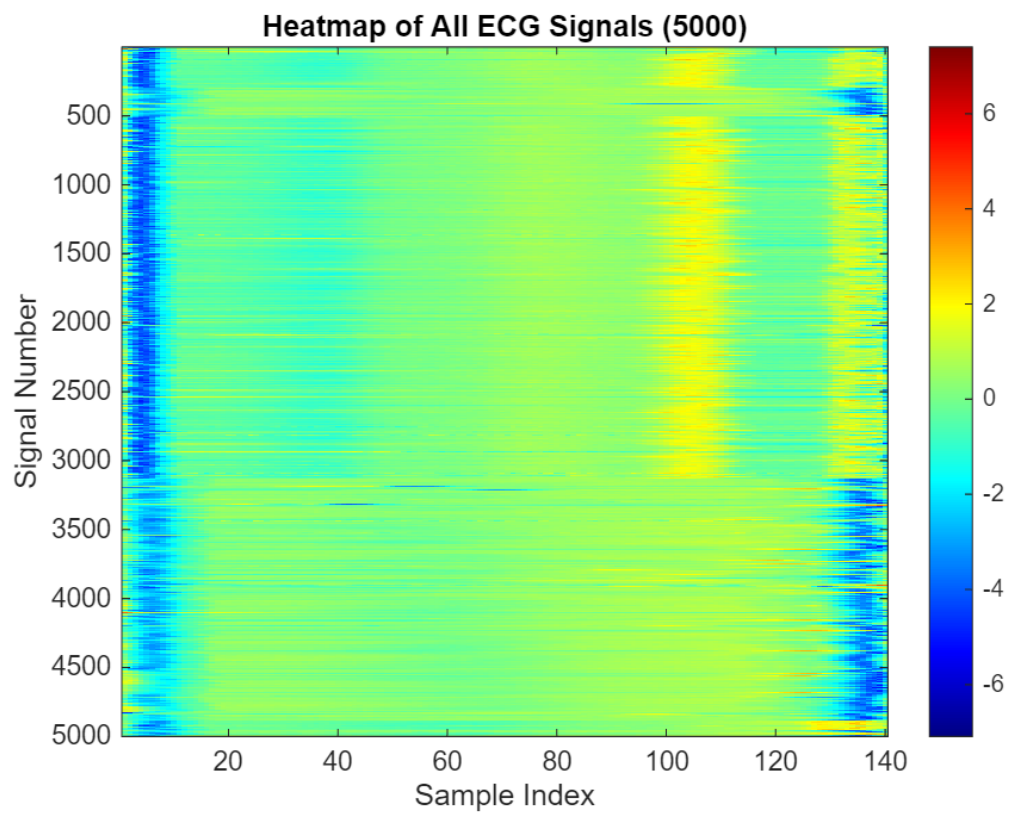
867	9
-----	---

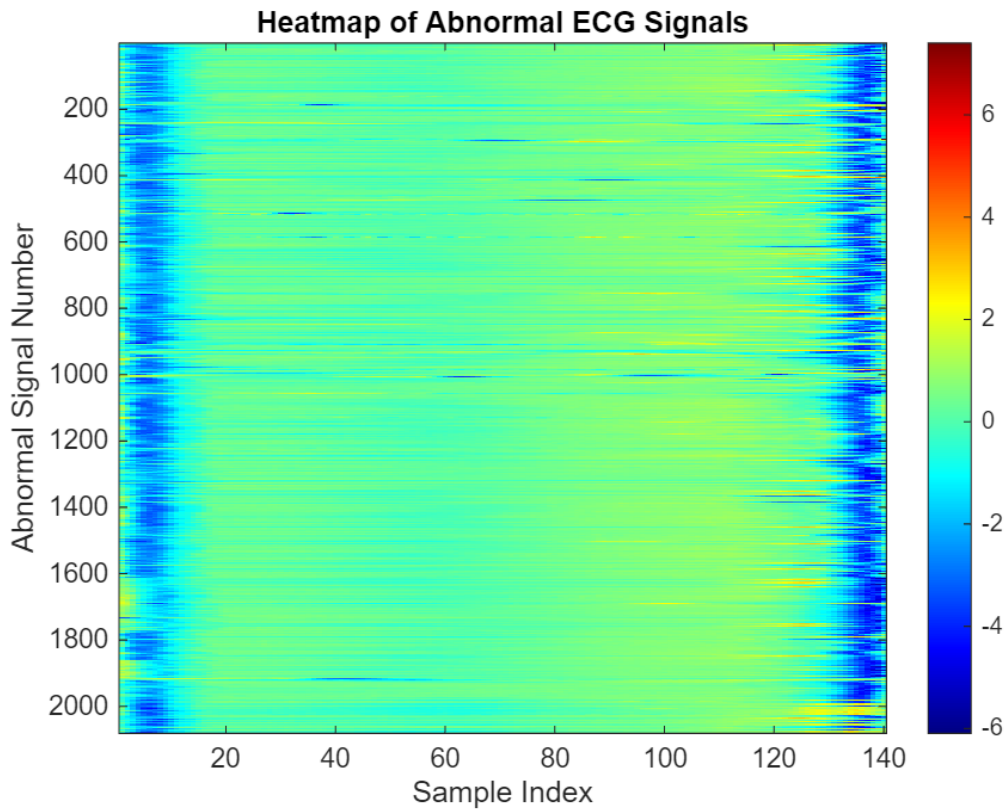
45	579
----	-----



1 SIGNAL EXAMPLE :







CONCLUSION

This project demonstrates the use of adaptive Toeplitz convolution for analyzing ECG signals and detecting abnormalities. The ECG5000 dataset was used to obtain real heart signal recordings, which were then normalized and processed using an adaptive convolution kernel. By constructing a Toeplitz matrix and performing convolution, important signal characteristics were enhanced, allowing abnormal patterns in ECG signals to become more distinguishable.

Statistical features such as mean, standard deviation, signal range, skewness, kurtosis, and RMS were extracted from the processed signals to represent the signal behavior numerically. These features were used to train a K-Nearest Neighbors (KNN) classifier that distinguishes between normal and abnormal ECG signals. The experimental results show that the proposed approach achieves high classification accuracy and good discrimination capability, as demonstrated by the confusion matrix and ROC curve analysis.

Overall, the combination of signal processing and machine learning provides an efficient framework for automated ECG signal analysis. The adaptive Toeplitz convolution method improves feature extraction and highlights abnormal patterns in ECG signals, making it useful for medical signal processing applications and automated cardiac monitoring systems.

ACKNOWLEDGEMENT

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